Victor Hui

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Education

University of California, Berkeley **Expected Graduation: May 2027**

Double Major: Bachelor of Arts in Data Science and Applied Mathematics *Berkeley, CA*

Relevant Coursework: Principles and Techniques of Data Science, Data Structures and Algorithms, Structures and Interpretation of Computer Programs, Efficient Algorithms and Intractable Problems, Probability and Random Processes, Linear Algebra and Differential Equations, Web Development, Advanced Programming in R, Intro to ML, Intro to AI

Experience

Berkeley Division of Undergraduate Education **Jan 2026 – Present**

Data Analyst *Berkeley, CA*

- Developed **Tableau** dashboards and automated reports summarizing survey responses from 10,000+ undergraduates, providing insights directly to the **Chancellor** and **Vice Provost** on academic programs and undergraduate courses
- Engineered and validated data sources from 100,000+ student records and survey responses, performing EDA to ensure high-quality, accurate analysis that powered dashboards and guided initiatives affecting 30,000+ students
- Influenced key decisions for 7+ campus departments by providing analysis representing the voices of all undergraduates

Certis Oncology **Aug 2025 – Jan 2026**

Software Developer (PM, Tech Lead, Contract) *Berkeley, CA*

- Led 13 engineers/designers to integrate an LLM-based prediction pipeline into a clinical data platform, enabling natural language querying and structured insight generation over tumor model data for research workflows across 10+ clients
- Built scalable data pipelines with **Python (Pandas)** and **PostgreSQL**, using per-user schemas and multi-tenant isolation to ingest, clean, and analyze tumor metadata across multiple clients and large datasets
- Improved system reliability through prompt tuning, offline testing, and user feedback, supporting real-time interaction via **WebSockets/SSE (Angular, NestJS)** for secure data exploration and multi-threaded user sessions

BLCK UNICRN **Jan 2025 – Jun 2025**

Machine Learning Researcher *Berkeley, CA*

- Researched and implemented clustering algorithms (**K-Means**) with **Python** and **SQL** on **AWS** to segment 1,200+ users, performing feature engineering and dimensionality reduction to uncover behavioral groups
- Developed regression and classification models with **scikit-learn** and **TensorFlow**, applying cross-validation, hyperparameter tuning, and A/B testing to predict subscription behavior and customer lifetime value
- Automated pipelines and dashboards to track model metrics (AUC, precision/recall, RMSE), improving retention by 8%

Recidiviz **Jan 2025 – May 2025**

Software Developer (Contract) *Berkeley, CA*

- Engineered an LLM-powered SMS check-in system using **TypeScript, LangChain**, and **Twilio**, processing 100+ weekly patrol officer case notes into personalized messages, reducing manual drafting time by 60% while flagging urgent needs
- Designed and deployed scalable data pipelines on **Firebase**, implementing topic schemas, confidence scoring, and scheduling logic to prioritize high-risk cases and reduce officer follow-up delays by 40%
- Built full-stack officer tools (case note UI, chat logs, evaluation framework, Google Calendar scheduling), enabling review of 200+ client interactions per month with structured evaluation workflows and improved LLM output reliability

Mindful Quality **May 2024 – Sep 2024**

Machine Learning Engineer Intern *Berkeley, CA*

- Developed a pharmaceutical based LLM using **PyTorch** and **Transformers** with 85% accuracy, using EDA, feature engineering, and hyperparameter tuning to optimize performance, improving analytics accuracy by 30%
- Researched and fine-tuned NLP models for document classification and report generation, optimizing **T5/BART** for text generation, while implementing custom tokenization for compliance automation for document accuracy
- Improved the company's overall capacity to efficiently serve clients in the US and India by over 20%

Projects

Spam Email Classification | *Python*

- Built a logistic regression model using **scikit-learn** to classify spam emails achieving 92% accuracy on validation data
- Performed EDA with **Matplotlib/Seaborn** and iterative feature engineering with **Pandas** on text features and encoded categorical variables to identify high-impact predictors, optimizing recall for spam emails by 10%
- Evaluated model performance using precision, recall, and ROC curve analysis, reducing false positives by 15%

Additional Skills

Programming Languages: Python, Java, C/C++, SQL, R, JavaScript (TypeScript), HTML/CSS
Developer Tools: Tableau, Jupyter Notebook, RStudio, Git, VS Code, Google Cloud Platform (GCP), AWS (S3)
Technologies/Frameworks: React.js, Node.js, Flask, PostgreSQL, Snowflake, Firebase, Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, TensorFlow, PyTorch, Transformers (T5, BART), LangChain, Twilio, Angular, NestJS